

Task

Perform an exploratory data analysis of the `/content/sample_full.csv` dataset, focusing on learning outcomes, including visualizations with legends, and summarize the key findings.

Load Dataset

Subtask:

Load the `/content/sample_full.csv` dataset into a pandas DataFrame for initial inspection.

Reasoning: Import the pandas library and load the specified CSV file into a DataFrame, then display the first few rows for verification.

```
In [1]: import os
import pandas as pd

CSV_PATH = "content/sample_full.csv" # apni CSV ka exact naam do
assert os.path.exists(CSV_PATH), f"CSV file not found: {CSV_PATH}"

df = pd.read_csv(CSV_PATH)
print("Dataset size:", len(df))
df.head()
```

Out[1]: Dataset size: 21380

Out[1]:

	Learning_outcome	Remember	Understand	Apply	Analyze	Evaluate	Create
0	Analyze the health economic implications of e...	NaN	NaN	NaN	1.0	NaN	NaN
1	Apply research skills to operate effectively ...	NaN	NaN	1.0	NaN	NaN	NaN
2	Assess and synthesise diverse information abo...	NaN	NaN	NaN	NaN	1.0	1.0
3	Describe the general characteristics of the m...	NaN	1.0	NaN	NaN	NaN	NaN
4	Evaluate the different models of perioperativ...	NaN	NaN	NaN	NaN	1.0	NaN

```
In [2]: print('Missing values per column:')
print(df.isnull().sum())
```

Out[2]: Missing values per column:

```
Learning_outcome      0
Remember              20195
Understand            15555
Apply                15299
Analyze              17921
Evaluate             17546
Create               17493
dtype: int64
```

Initial Data Exploration

Subtask:

Perform initial data exploration to understand the structure, data types, identify missing values, and get a summary of descriptive statistics.

Reasoning: To perform initial data exploration, I will use `.info()` to get a summary of the DataFrame's structure and data types, `.isnull().sum()` to identify missing values in each column, and `.describe()` to obtain descriptive statistics for numerical columns.

```
In [3]: print(df.info())
print("\nMissing values per column:")
print(df.isnull().sum())
print("\nDescriptive statistics:")
print(df.describe())
```

```
Out[3]: <class 'pandas.core.frame.DataFrame'>
RangeIndex: 21380 entries, 0 to 21379
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Learning_outcome 21380 non-null  object
 1   Remember         1185 non-null   float64
 2   Understand       5825 non-null   float64
 3   Apply           6081 non-null   float64
 4   Analyze         3459 non-null   float64
 5   Evaluate        3834 non-null   float64
 6   Create          3887 non-null   float64
dtypes: float64(6), object(1)
memory usage: 1.1+ MB
None
```

```
Missing values per column:
Learning_outcome      0
Remember              20195
Understand            15555
Apply                15299
Analyze              17921
Evaluate             17546
Create               17493
```

dtype: int64

Descriptive statistics:

	Remember	Understand	Apply	Analyze	Evaluate	Create
count	1185.0	5825.0	6081.0	3459.0	3834.0	3887.0
mean	1.0	1.0	1.0	1.0	1.0	1.0
std	0.0	0.0	0.0	0.0	0.0	0.0
min	1.0	1.0	1.0	1.0	1.0	1.0
25%	1.0	1.0	1.0	1.0	1.0	1.0
50%	1.0	1.0	1.0	1.0	1.0	1.0
75%	1.0	1.0	1.0	1.0	1.0	1.0
max	1.0	1.0	1.0	1.0	1.0	1.0

Analyze Learning Outcomes

Subtask:

Investigate the columns related to 'Learning Outcomes' to understand their distribution and unique values, and identify any patterns or categories.

Reasoning: To investigate the 'Learning_outcome' column and cognitive domain columns, I will first get the count of unique learning outcomes and display some samples. Then, for each cognitive domain column, I will sum the non-null values to determine the number of learning outcomes associated with each domain.

```
In [4]: print(f"Number of unique learning outcomes: {df['Learning_outcome'].nunique()}")
print("\nSample unique learning outcomes:")
print(df['Learning_outcome'].unique()[:5])

cognitive_domains = ['Remember', 'Understand', 'Apply', 'Analyze', 'Evaluate', 'Create']
cognitive_domain_counts = df[cognitive_domains].count()

print("\nLearning outcomes per cognitive domain:")
print(cognitive_domain_counts)
```

```
Out[4]: Number of unique learning outcomes: 21246
```

Sample unique learning outcomes:

```
[' Analyze the health economic implications of efficient perioperative practice. '
 ' Apply research skills to operate effectively as a member of a research project team. '
 ' Assess and synthesise diverse information about information and knowledge management
technologies market and how to use implementation strategies to maximise their strengths and
minimise their weaknesses. '
 ' Describe the general characteristics of the modern X-ray system used in clinical practice,
including scientific principles, and production of the digital image. '
 ' Evaluate the different models of perioperative care that exist globally. ']
```

Learning outcomes per cognitive domain:

```
Remember      1185
```

```
Understand    5825
Apply         6081
Analyze       3459
Evaluate      3834
Create        3887
dtype: int64
```

Visualize Key Metrics

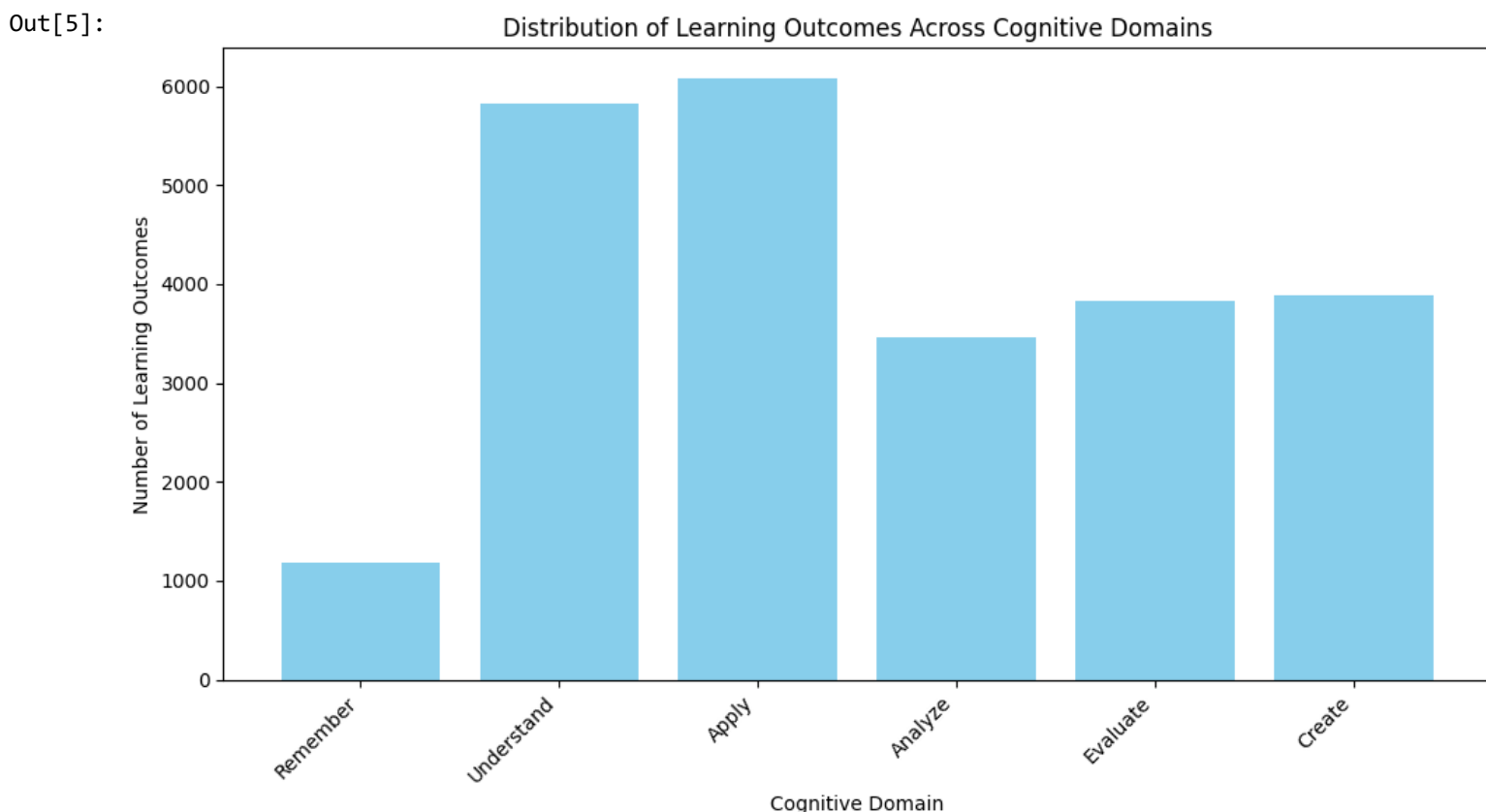
Subtask:

Generate visualizations, such as bar charts or histograms, to illustrate the distribution of learning outcomes or other relevant categorical/numerical data, including legends for clarity.

Reasoning: To visualize the distribution of learning outcomes across cognitive domains, I will create a bar chart using `matplotlib.pyplot` and label it appropriately for clarity.

```
In [5]: import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
plt.bar(cognitive_domain_counts.index, cognitive_domain_counts.values, color='skyblue')
plt.title('Distribution of Learning Outcomes Across Cognitive Domains')
plt.xlabel('Cognitive Domain')
plt.ylabel('Number of Learning Outcomes')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Final Task

Subtask:

Summarize the key findings from the exploratory data analysis of the Moonash University learning outcomes dataset.

Summary:

Data Analysis Key Findings

- The dataset contains 21,380 entries and 7 columns, including `Learning_outcome` (object type) and six cognitive domain columns (`Remember`, `Understand`, `Apply`, `Analyze`, `Evaluate`, `Create`) which are `float64` type.
- The `Learning_outcome` column has no missing values, but all cognitive domain columns have a very high number of missing values, ranging from 15,299 (`Apply`) to 20,195 (`Remember`), indicating significant data sparsity.
- For non-null entries, the values in the cognitive domain columns are uniformly 1.0 with a standard deviation of 0.0, suggesting they act as binary indicators (1.0 or NaN) rather than continuous variables.
- There are 21,246 unique entries in the `Learning_outcome` column, indicating a high diversity of learning outcomes.
- The distribution of learning outcomes across cognitive domains shows:
 - `Remember`: 1,185
 - `Understand`: 5,825
 - `Apply`: 6,081
 - `Analyze`: 3,459
 - `Evaluate`: 3,834
 - `Create`: 3,887
 - `Apply` (6,081) and `Understand` (5,825) are the most frequently associated cognitive domains, while `Remember` (1,185) has the fewest associations.
- A bar chart visually confirmed that 'Apply' and 'Understand' domains have the highest counts of associated learning outcomes, and 'Remember' has the lowest.

Insights or Next Steps

- Given the binary nature of the cognitive domain columns and the high proportion of missing values, further investigation is needed to understand if 'NaN' implies the absence of that cognitive domain or genuinely missing data. This will influence how these columns are used for analysis.
- Due to the large number of unique `Learning_outcome` entries, natural language processing (NLP) techniques could be applied to categorize or cluster similar learning outcomes, providing deeper insights into the curriculum structure.

Bloom Taxonomy Classification using Open Source LLMs

This notebook evaluates open-source Large Language Models on the **Monash University Learning Outcomes dataset**.

Objective

Classify learning outcomes into Bloom's Taxonomy levels:

- Remember
- Understand
- Apply
- Analyze
- Evaluate
- Create

Approach

1. Load dataset
2. Convert one-hot labels to single label
3. Split dataset
4. Prompt LLM for classification
5. Evaluate predictions

Models tested:

- Mistral 7B Instruct
- Gemma 2B IT
- Llama 3 8B Instruct

◆ 1. Environment Setup

```
In [6]: %pip install -U pandas numpy scikit-learn matplotlib tqdm datasets evaluate
        %pip install -U transformers accelerate sentencepiece
        %pip install -U bitsandbytes
        %pip uninstall torch torchvision torchaudio -y
        %pip uninstall nvidia-* -y
        %pip uninstall nvidia-cuda-runtime-cu12 nvidia-cublas-cu12 nvidia-cusparse-cu12 -y
        %pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118
```

```
Out[6]: Requirement already satisfied: pandas in /usr/local/lib/python3.11/dist-packages (2.2.0)
        Collecting pandas
```

```
Downloading pandas-3.0.2-cp311-cp311-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)
79.5/79.5 kB 2.3 MB/s eta 0:00:00
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (1.26.3)
Collecting numpy
  Downloading numpy-2.4.4-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.11/dist-packages
(1.3.0)
Collecting scikit-learn
  Downloading scikit_learn-1.8.0-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (3.7.3)
Collecting matplotlib
  Downloading matplotlib-3.10.8-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (52 kB)
52.8/52.8 kB 12.7 MB/s eta 0:00:00
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (4.66.1)
Collecting tqdm
  Downloading tqdm-4.67.3-py3-none-any.whl.metadata (57 kB)
57.7/57.7 kB 14.9 MB/s eta 0:00:00
Requirement already satisfied: datasets in /usr/local/lib/python3.11/dist-packages (2.14.5)
Collecting datasets
  Downloading datasets-4.8.4-py3-none-any.whl.metadata (19 kB)
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-
packages (from pandas) (2.8.2)
Requirement already satisfied: scipy>=1.10.0 in /usr/local/lib/python3.11/dist-packages (from
scikit-learn) (1.11.2)
Requirement already satisfied: joblib>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from
scikit-learn) (1.3.2)
Requirement already satisfied: threadpoolctl>=3.2.0 in /usr/local/lib/python3.11/dist-
packages (from scikit-learn) (3.2.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (1.2.0)
Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.11/dist-packages (from
matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (4.47.2)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (1.4.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages
(from matplotlib) (23.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (from
matplotlib) (9.5.0)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from
datasets) (3.13.1)
Collecting pyarrow>=21.0.0 (from datasets)
  Downloading pyarrow-23.0.1-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (3.1 kB)
Requirement already satisfied: dill<0.4.2,>=0.3.0 in /usr/local/lib/python3.11/dist-packages
(from datasets) (0.3.7)
Collecting requests>=2.32.2 (from datasets)
```

Downloading requests-2.33.1-py3-none-any.whl.metadata (4.8 kB)
Collecting httpx<1.0.0 (from datasets)
Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: xxhash in /usr/local/lib/python3.11/dist-packages (from datasets) (3.4.1)
Requirement already satisfied: multiprocessing<0.70.20 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.70.15)
Requirement already satisfied: fsspec<=2026.2.0,>=2023.1.0 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (2023.6.0)
Collecting huggingface-hub<2.0,>=0.25.0 (from datasets)
Downloading huggingface-hub-1.8.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: pyyaml>=5.1 in /usr/lib/python3/dist-packages (from datasets) (5.4.1)
Collecting numpy
Downloading numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)

61.0/61.0 kB 17.9 MB/s eta 0:00:00
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (3.9.1)
Requirement already satisfied: anyio in /usr/local/lib/python3.11/dist-packages (from httpx<1.0.0->datasets) (4.2.0)
Requirement already satisfied: certifi in /usr/lib/python3/dist-packages (from httpx<1.0.0->datasets) (2020.6.20)
Collecting httpcore==1.* (from httpx<1.0.0->datasets)
Downloading httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Requirement already satisfied: idna in /usr/lib/python3/dist-packages (from httpx<1.0.0->datasets) (3.3)
Collecting h11>=0.16 (from httpcore==1.*->httpx<1.0.0->datasets)
Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting hf-xet<2.0.0,>=1.4.2 (from huggingface-hub<2.0,>=0.25.0->datasets)
Downloading hf-xet-1.4.2-cp37-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (4.9 kB)
Requirement already satisfied: typer in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (0.9.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (4.9.0)
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (3.3.2)
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2.0.7)
Collecting certifi (from httpx<1.0.0->datasets)
Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (23.1.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (6.0.4)
Requirement already satisfied: yarl<2.0,>=1.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.9.4)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.4.1)
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.3.1)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/dist-packages (from

```

anyio->httpx<1.0.0->datasets) (1.3.0)
Requirement already satisfied: click<9.0.0,>=7.1.1 in /usr/local/lib/python3.11/dist-packages
(from typer->huggingface-hub<2.0,>=0.25.0->datasets) (8.1.7)
Downloading pandas-3.0.2-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (11.3
MB)
_____ 11.3/11.3 MB 142.9 MB/s eta 0:00:0000:010:01
Downloading scikit_learn-1.8.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
(9.1 MB)
_____ 9.1/9.1 MB 123.1 MB/s eta 0:00:0000:0100:01
Downloading matplotlib-3.10.8-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.7
MB)
_____ 8.7/8.7 MB 165.6 MB/s eta 0:00:0000:01
Downloading tqdm-4.67.3-py3-none-any.whl (78 kB)
_____ 78.4/78.4 kB 24.1 MB/s eta 0:00:00
Downloading datasets-4.8.4-py3-none-any.whl (526 kB)
_____ 527.0/527.0 kB 68.9 MB/s eta 0:00:00
Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
_____ 84.1/84.1 kB 27.0 MB/s eta 0:00:00
Downloading numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.3 MB)
_____ 18.3/18.3 MB 97.3 MB/s eta 0:00:00:00:0100:01
Downloading httpx-0.28.1-py3-none-any.whl (73 kB)
_____ 73.5/73.5 kB 19.8 MB/s eta 0:00:00
Downloading httpcore-1.0.9-py3-none-any.whl (78 kB)
_____ 78.8/78.8 kB 23.8 MB/s eta 0:00:00
Downloading huggingface_hub-1.8.0-py3-none-any.whl (625 kB)
_____ 625.2/625.2 kB 100.3 MB/s eta 0:00:00
Downloading pyarrow-23.0.1-cp311-cp311-manylinux_2_28_x86_64.whl (47.6 MB)
_____ 47.6/47.6 MB 64.2 MB/s eta 0:00:00:00:0100:01
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
_____ 122.8/122.8 kB 34.9 MB/s eta 0:00:00
Downloading requests-2.33.1-py3-none-any.whl (64 kB)
_____ 64.9/64.9 kB 22.1 MB/s eta 0:00:00
Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)
_____ 153.7/153.7 kB 45.7 MB/s eta 0:00:00
Downloading hf_xet-1.4.2-cp37-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (4.2 MB)
_____ 4.2/4.2 MB 140.2 MB/s eta 0:00:00
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Installing collected packages: tqdm, pyparsing, pyarrow, numpy, hf-xet, h11, certifi,
requests, pandas, httpcore, scikit-learn, matplotlib, httpx, huggingface-hub, datasets,
evaluate
  Attempting uninstall: tqdm
    Found existing installation: tqdm 4.66.1
    Uninstalling tqdm-4.66.1:
      Successfully uninstalled tqdm-4.66.1
  Attempting uninstall: pyparsing
    Found existing installation: pyparsing 2.4.7
    Uninstalling pyparsing-2.4.7:
      Successfully uninstalled pyparsing-2.4.7
  Attempting uninstall: pyarrow
    Found existing installation: pyarrow 15.0.0
    Uninstalling pyarrow-15.0.0:
      Successfully uninstalled pyarrow-15.0.0
  Attempting uninstall: numpy
    Found existing installation: numpy 1.26.3
    Uninstalling numpy-1.26.3:

```

```

Successfully uninstalled numpy-1.26.3
Attempting uninstall: certifi
  Found existing installation: certifi 2020.6.20
  Uninstalling certifi-2020.6.20:
    Successfully uninstalled certifi-2020.6.20
Attempting uninstall: requests
  Found existing installation: requests 2.31.0
  Uninstalling requests-2.31.0:
    Successfully uninstalled requests-2.31.0
Attempting uninstall: pandas
  Found existing installation: pandas 2.2.0
  Uninstalling pandas-2.2.0:
    Successfully uninstalled pandas-2.2.0
Attempting uninstall: scikit-learn
  Found existing installation: scikit-learn 1.3.0
  Uninstalling scikit-learn-1.3.0:
    Successfully uninstalled scikit-learn-1.3.0
Attempting uninstall: matplotlib
  Found existing installation: matplotlib 3.7.3
  Uninstalling matplotlib-3.7.3:
    Successfully uninstalled matplotlib-3.7.3
Attempting uninstall: huggingface-hub
  Found existing installation: huggingface-hub 0.20.3
  Uninstalling huggingface-hub-0.20.3:
    Successfully uninstalled huggingface-hub-0.20.3
Attempting uninstall: datasets
  Found existing installation: datasets 2.14.5
  Uninstalling datasets-2.14.5:
    Successfully uninstalled datasets-2.14.5
ERROR: pip's dependency resolver does not currently take into account all the packages that
are installed. This behaviour is the source of the following dependency conflicts.
gradient 2.0.6 requires attrs<=19, but you have attrs 23.1.0 which is incompatible.
transformers 4.35.2 requires huggingface-hub<1.0,>=0.16.4, but you have huggingface-hub 1.8.0
which is incompatible.
tokenizers 0.15.1 requires huggingface_hub<1.0,>=0.16.4, but you have huggingface-hub 1.8.0
which is incompatible.
Successfully installed certifi-2026.2.25 datasets-4.8.4 evaluate-0.4.6 h11-0.16.0 hf-
xet-1.4.2 httpcore-1.0.9 httpx-0.28.1 huggingface-hub-1.8.0 matplotlib-3.10.8 numpy-1.26.4
pandas-3.0.2 pyarrow-23.0.1 pyparsing-3.3.2 requests-2.33.1 scikit-learn-1.8.0 tqdm-4.67.3
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting
behaviour with the system package manager. It is recommended to use a virtual environment
instead: https://pip.pypa.io/warnings/venv
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: transformers in /usr/local/lib/python3.11/dist-packages
(4.35.2)
Collecting transformers
  Downloading transformers-5.4.0-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.11/dist-packages (0.24.1)
Collecting accelerate
  Downloading accelerate-1.13.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.11/dist-packages
(0.1.99)
Collecting sentencepiece
  Downloading sentencepiece-0.2.1-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (10 kB)

```

Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (1.8.0)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from transformers) (1.26.4)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (23.2)

Requirement already satisfied: pyyaml>=5.1 in /usr/lib/python3/dist-packages (from transformers) (5.4.1)

Collecting regex>=2025.10.22 (from transformers)

 Downloading regex-2026.3.32-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (40 kB)

40.4/40.4 kB 7.3 MB/s eta 0:00:00

Collecting tokenizers<=0.23.0,>=0.22.0 (from transformers)

 Downloading tokenizers-0.22.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.3 kB)

Requirement already satisfied: typer in /usr/local/lib/python3.11/dist-packages (from transformers) (0.9.0)

Collecting safetensors>=0.4.3 (from transformers)

 Downloading safetensors-0.7.0-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)

Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.11/dist-packages (from transformers) (4.67.3)

Requirement already satisfied: psutil in /usr/local/lib/python3.11/dist-packages (from accelerate) (5.9.8)

Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from accelerate) (2.1.1+cu121)

Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3.13.1)

Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2023.6.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.4.2)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.28.1)

Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (4.9.0)

Requirement already satisfied: sympy in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (1.12)

Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (3.2.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (3.1.3)

Requirement already satisfied: triton==2.1.0 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (2.1.0)

Requirement already satisfied: click<9.0.0,>=7.1.1 in /usr/local/lib/python3.11/dist-packages (from typer->transformers) (8.1.7)

Requirement already satisfied: anyio in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (4.2.0)

Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (2026.2.25)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.9)

Requirement already satisfied: idna in /usr/lib/python3/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (3.3)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.11/dist-packages (from

httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages
(from jinja2->torch>=2.0.0->accelerate) (2.1.4)
Requirement already satisfied: mpmath>=0.19 in /usr/local/lib/python3.11/dist-packages (from
sympy->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/dist-packages (from
anyio->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.3.0)
Downloading transformers-5.4.0-py3-none-any.whl (10.1 MB)

10.1/10.1 MB 59.2 MB/s eta 0:00:00 0:00:01m

Downloading accelerate-1.13.0-py3-none-any.whl (383 kB)

383.7/383.7 kB 82.0 MB/s eta 0:00:00

Downloading sentencepiece-0.2.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
(1.4 MB)

1.4/1.4 MB 123.2 MB/s eta 0:00:00

Downloading regex-2026.3.32-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (798 kB)

798.3/798.3 kB 114.6 MB/s eta 0:00:00

Downloading safetensors-0.7.0-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (507
kB)

507.2/507.2 kB 93.8 MB/s eta 0:00:00

Downloading tokenizers-0.22.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.3
MB)

3.3/3.3 MB 142.4 MB/s eta 0:00:00

Installing collected packages: sentencepiece, safetensors, regex, tokenizers, accelerate,
transformers

Attempting uninstall: sentencepiece

Found existing installation: sentencepiece 0.1.99

Uninstalling sentencepiece-0.1.99:

Successfully uninstalled sentencepiece-0.1.99

Attempting uninstall: safetensors

Found existing installation: safetensors 0.4.0

Uninstalling safetensors-0.4.0:

Successfully uninstalled safetensors-0.4.0

Attempting uninstall: regex

Found existing installation: regex 2023.12.25

Uninstalling regex-2023.12.25:

Successfully uninstalled regex-2023.12.25

Attempting uninstall: tokenizers

Found existing installation: tokenizers 0.15.1

Uninstalling tokenizers-0.15.1:

Successfully uninstalled tokenizers-0.15.1

Attempting uninstall: accelerate

Found existing installation: accelerate 0.24.1

Uninstalling accelerate-0.24.1:

Successfully uninstalled accelerate-0.24.1

Attempting uninstall: transformers

Found existing installation: transformers 4.35.2

Uninstalling transformers-4.35.2:

Successfully uninstalled transformers-4.35.2

ERROR: pip's dependency resolver does not currently take into account all the packages that
are installed. This behaviour is the source of the following dependency conflicts.
sentence-transformers 2.2.2 requires transformers<5.0.0,>=4.6.0, but you have transformers
5.4.0 which is incompatible.

Successfully installed accelerate-1.13.0 regex-2026.3.32 safetensors-0.7.0
sentencepiece-0.2.1 tokenizers-0.22.2 transformers-5.4.0

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: bitsandbytes in /usr/local/lib/python3.11/dist-packages (0.41.2)

Collecting bitsandbytes

Downloading bitsandbytes-0.49.2-py3-none-manylinux_2_24_x86_64.whl.metadata (10 kB)

Collecting torch<3,>=2.3 (from bitsandbytes)

Downloading torch-2.11.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (29 kB)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from bitsandbytes) (1.26.4)

Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.11/dist-packages (from bitsandbytes) (23.2)

Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (3.13.1)

Collecting typing-extensions>=4.10.0 (from torch<3,>=2.3->bitsandbytes)

Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (69.0.3)

Collecting sympy>=1.13.3 (from torch<3,>=2.3->bitsandbytes)

Downloading sympy-1.14.0-py3-none-any.whl.metadata (12 kB)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (3.2.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (3.1.3)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.3->bitsandbytes) (2023.6.0)

Collecting cuda-toolkit==13.0.2 (from cuda-

toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)

Downloading cuda_toolkit-13.0.2-py2.py3-none-any.whl.metadata (9.4 kB)

Collecting cuda-bindings<14,>=13.0.3 (from torch<3,>=2.3->bitsandbytes)

Downloading cuda_bindings-13.2.0-cp311-cp311-

manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (2.3 kB)

Collecting nvidia-cudnn-cu13==9.19.0.56 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_cudnn_cu13-9.19.0.56-py3-none-manylinux_2_27_x86_64.whl.metadata (1.9 kB)

Collecting nvidia-cusparselt-cu13==0.8.0 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_cusparselt_cu13-0.8.0-py3-none-manylinux2014_x86_64.whl.metadata (12 kB)

Collecting nvidia-nccl-cu13==2.28.9 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_nccl_cu13-2.28.9-py3-none-manylinux_2_18_x86_64.whl.metadata (2.0 kB)

Collecting nvidia-nvshmem-cu13==3.4.5 (from torch<3,>=2.3->bitsandbytes)

Downloading nvidia_nvshmem_cu13-3.4.5-py3-none-

manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.1 kB)

Collecting triton==3.6.0 (from torch<3,>=2.3->bitsandbytes)

Downloading triton-3.6.0-cp311-cp311-

manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.7 kB)

Collecting nvidia-cublas==13.1.0.3.* (from cuda-

toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)

Downloading nvidia_cublas-13.1.0.3-py3-none-manylinux_2_27_x86_64.whl.metadata (1.7 kB)

Collecting nvidia-cuda-runtime==13.0.96.* (from cuda-

toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.2 platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)

```
Downloading nvidia_cuda_runtime-13.0.96-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cufft==12.0.0.61.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_cufft-12.0.0.61-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cufile==1.15.1.6.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_cufile-1.15.1.6-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-cupti==13.0.85.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_cuda_cupti-13.0.85-py3-none-manylinux_2_25_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-curand==10.4.0.35.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_curand-10.4.0.35-py3-none-manylinux_2_27_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cusolver==12.0.4.66.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_cusolver-12.0.4.66-py3-none-manylinux_2_27_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cuspars
e==12.6.3.3.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_cuspars
e-12.6.3.3-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-nvjitlink==13.0.88.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_nvjitlink-13.0.88-py3-none-
manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-nvrtc==13.0.88.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-
manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-nvtx==13.0.85.* (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==13.0.2
platform_system == "Linux"->torch<3,>=2.3->bitsandbytes)
  Downloading nvidia_nvtx-13.0.85-py3-none-
manylinux1_x86_64.manylinux_2_5_x86_64.whl.metadata (1.8 kB)
Collecting cuda-pathfinder~=1.1 (from cuda-bindings<14,>=13.0.3->torch<3,>=2.3->bitsandbytes)
  Downloading cuda_pathfinder-1.5.0-py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages
(from sympy>=1.13.3->torch<3,>=2.3->bitsandbytes) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages
(from jinja2->torch<3,>=2.3->bitsandbytes) (2.1.4)
Downloading bitsandbytes-0.49.2-py3-none-manylinux_2_24_x86_64.whl (60.7 MB)
_____ 60.7/60.7 MB 50.7 MB/s eta 0:00:00:00:0100:01
Downloading torch-2.11.0-cp311-cp311-manylinux_2_28_x86_64.whl (530.6 MB)
_____ 530.6/530.6 MB 8.0 MB/s eta 0:00:00:00:0100:01
Downloading cuda_toolkit-13.0.2-py2.py3-none-any.whl (2.4 kB)
```

```
Downloading nvidia_cudnn_cu13-9.19.0.56-py3-none-manylinux_2_27_x86_64.whl (366.1 MB)
_____ 366.1/366.1 MB 10.6 MB/s eta 0:00:000:0100:01
Downloading nvidia_cusparselt_cu13-0.8.0-py3-none-manylinux2014_x86_64.whl (169.9 MB)
_____ 169.9/169.9 MB 21.8 MB/s eta 0:00:000:0100:01
Downloading nvidia_nccl_cu13-2.28.9-py3-none-manylinux_2_18_x86_64.whl (196.5 MB)
_____ 196.5/196.5 MB 17.6 MB/s eta 0:00:000:0100:01
Downloading nvidia_nvshmem_cu13-3.4.5-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
(60.4 MB)
_____ 60.4/60.4 MB 45.5 MB/s eta 0:00:00:00:0100:01
Downloading triton-3.6.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (188.2
MB)
_____ 188.2/188.2 MB 19.1 MB/s eta 0:00:000:0100:01
Downloading nvidia_cublas-13.1.0.3-py3-none-manylinux_2_27_x86_64.whl (423.1 MB)
_____ 423.1/423.1 MB 9.2 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_cuda_cupti-13.0.85-py3-none-manylinux_2_25_x86_64.whl (10.7 MB)
_____ 10.7/10.7 MB 133.8 MB/s eta 0:00:000:010:01
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl
(90.2 MB)
_____ 90.2/90.2 MB 33.5 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_cuda_runtime-13.0.96-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl (2.2 MB)
_____ 2.2/2.2 MB 125.7 MB/s eta 0:00:00
Downloading nvidia_cufft-12.0.0.61-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
(214.1 MB)
_____ 214.1/214.1 MB 18.8 MB/s eta 0:00:000:0100:01
Downloading nvidia_cufile-1.15.1.6-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
(1.2 MB)
_____ 1.2/1.2 MB 109.3 MB/s eta 0:00:00
Downloading nvidia_curand-10.4.0.35-py3-none-manylinux_2_27_x86_64.whl (59.5 MB)
_____ 59.5/59.5 MB 54.9 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_cusolver-12.0.4.66-py3-none-manylinux_2_27_x86_64.whl (200.9 MB)
_____ 200.9/200.9 MB 18.1 MB/s eta 0:00:000:0100:01
Downloading nvidia_cusparselt-12.6.3.3-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
(145.9 MB)
_____ 145.9/145.9 MB 26.1 MB/s eta 0:00:000:0100:01
Downloading nvidia_nvjitlink-13.0.88-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl
(40.7 MB)
_____ 40.7/40.7 MB 70.4 MB/s eta 0:00:00:00:0100:01
Downloading nvidia_nvtx-13.0.85-py3-none-manylinux1_x86_64.manylinux_2_5_x86_64.whl (148 kB)
_____ 148.0/148.0 kB 35.1 MB/s eta 0:00:00
Downloading cuda_bindings-13.2.0-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl
(6.3 MB)
_____ 6.3/6.3 MB 141.5 MB/s eta 0:00:000:00:01
Downloading sympy-1.14.0-py3-none-any.whl (6.3 MB)
_____ 6.3/6.3 MB 147.7 MB/s eta 0:00:000:00:01
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
_____ 44.6/44.6 kB 12.1 MB/s eta 0:00:00
Downloading cuda_pathfinder-1.5.0-py3-none-any.whl (49 kB)
_____ 49.7/49.7 kB 14.8 MB/s eta 0:00:00
```

Installing collected packages: nvidia-cusparselt-cu13, cuda-toolkit, typing-extensions, triton, sympy, nvidia-nvtx, nvidia-nvshmem-cu13, nvidia-nvjitlink, nvidia-nccl-cu13, nvidia-curand, nvidia-cufile, nvidia-cuda-runtime, nvidia-cuda-nvrtc, nvidia-cuda-cupti, nvidia-cublas, cuda-pathfinder, nvidia-cusparselt, nvidia-cufft, nvidia-cudnn-cu13, cuda-bindings, nvidia-cusolver, torch, bitsandbytes

Attempting uninstall: typing-extensions

```

Found existing installation: typing_extensions 4.9.0
Uninstalling typing_extensions-4.9.0:
  Successfully uninstalled typing_extensions-4.9.0
Attempting uninstall: triton
  Found existing installation: triton 2.1.0
  Uninstalling triton-2.1.0:
    Successfully uninstalled triton-2.1.0
Attempting uninstall: sympy
  Found existing installation: sympy 1.12
  Uninstalling sympy-1.12:
    Successfully uninstalled sympy-1.12
Attempting uninstall: torch
  Found existing installation: torch 2.1.1+cu121
  Uninstalling torch-2.1.1+cu121:
    Successfully uninstalled torch-2.1.1+cu121
Attempting uninstall: bitsandbytes
  Found existing installation: bitsandbytes 0.41.2
  Uninstalling bitsandbytes-0.41.2:
    Successfully uninstalled bitsandbytes-0.41.2
ERROR: pip's dependency resolver does not currently take into account all the packages that
are installed. This behaviour is the source of the following dependency conflicts.
sentence-transformers 2.2.2 requires transformers<5.0.0,>=4.6.0, but you have transformers
5.4.0 which is incompatible.
torchaudio 2.1.1+cu121 requires torch==2.1.1, but you have torch 2.11.0 which is
incompatible.
torchvision 0.16.1+cu121 requires torch==2.1.1, but you have torch 2.11.0 which is
incompatible.
Successfully installed bitsandbytes-0.49.2 cuda-bindings-13.2.0 cuda-pathfinder-1.5.0 cuda-
toolkit-13.0.2 nvidia-cublas-13.1.0.3 nvidia-cuda-cupti-13.0.85 nvidia-cuda-nvrtc-13.0.88
nvidia-cuda-runtime-13.0.96 nvidia-cudnn-cu13-9.19.0.56 nvidia-cufft-12.0.0.61 nvidia-
cufile-1.15.1.6 nvidia-curand-10.4.0.35 nvidia-cusolver-12.0.4.66 nvidia-cusparselt-12.6.3.3
nvidia-cusparselt-cu13-0.8.0 nvidia-nccl-cu13-2.28.9 nvidia-nvjitlink-13.0.88 nvidia-nvshmem-
cu13-3.4.5 nvidia-nvtx-13.0.85 sympy-1.14.0 torch-2.11.0 triton-3.6.0 typing-
extensions-4.15.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting
behaviour with the system package manager. It is recommended to use a virtual environment
instead: https://pip.pypa.io/warnings/venv
Note: you may need to restart the kernel to use updated packages.
Found existing installation: torch 2.11.0
Uninstalling torch-2.11.0:
  Successfully uninstalled torch-2.11.0
Found existing installation: torchvision 0.16.1+cu121
Uninstalling torchvision-0.16.1+cu121:
  Successfully uninstalled torchvision-0.16.1+cu121
Found existing installation: torchaudio 2.1.1+cu121
Uninstalling torchaudio-2.1.1+cu121:
  Successfully uninstalled torchaudio-2.1.1+cu121
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting
behaviour with the system package manager. It is recommended to use a virtual environment
instead: https://pip.pypa.io/warnings/venv
Note: you may need to restart the kernel to use updated packages.
ERROR: Invalid requirement: 'nvidia-*'
Note: you may need to restart the kernel to use updated packages.
WARNING: Skipping nvidia-cuda-runtime-cu12 as it is not installed.
WARNING: Skipping nvidia-cublas-cu12 as it is not installed.

```

WARNING: Skipping nvidia-cusparse-cu12 as it is not installed.
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

Note: you may need to restart the kernel to use updated packages.

Looking in indexes: <https://download.pytorch.org/whl/cu118>

Collecting torch

Downloading https://download.pytorch.org/whl/cu118/torch-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (28 kB)

Collecting torchvision

Downloading https://download-r2.pytorch.org/whl/cu118/torchvision-0.22.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (6.1 kB)

Collecting torchaudio

Downloading https://download-r2.pytorch.org/whl/cu118/torchaudio-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (6.6 kB)

Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch) (3.13.1)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.11/dist-packages (from torch) (4.15.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.11/dist-packages (from torch) (1.14.0)

Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch) (3.2.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch) (3.1.3)

Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from torch) (2023.6.0)

Collecting nvidia-cuda-nvrtc-cu11==11.8.89 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cuda_nvrtc_cu11-11.8.89-py3-none-manylinux1_x86_64.whl (23.2 MB)

23.2/23.2 MB 112.7 MB/s eta 0:00:000:0100:01

Collecting nvidia-cuda-runtime-cu11==11.8.89 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cuda_runtime_cu11-11.8.89-py3-none-manylinux1_x86_64.whl (875 kB)

875.6/875.6 kB 106.1 MB/s eta 0:00:00

Collecting nvidia-cuda-cupti-cu11==11.8.87 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cuda_cupti_cu11-11.8.87-py3-none-manylinux1_x86_64.whl (13.1 MB)

13.1/13.1 MB 137.6 MB/s eta 0:00:000:0100:01

Collecting nvidia-cudnn-cu11==9.1.0.70 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cudnn_cu11-9.1.0.70-py3-none-manylinux2014_x86_64.whl (663.9 MB)

663.9/663.9 MB 6.3 MB/s eta 0:00:000:0100:01

Collecting nvidia-cublas-cu11==11.11.3.6 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cublas_cu11-11.11.3.6-py3-none-manylinux1_x86_64.whl (417.9 MB)

417.9/417.9 MB 9.7 MB/s eta 0:00:000:0100:01

Collecting nvidia-cufft-cu11==10.9.0.58 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_cufft_cu11-10.9.0.58-py3-none-manylinux1_x86_64.whl (168.4 MB)

168.4/168.4 MB 21.3 MB/s eta 0:00:000:0100:01

Collecting nvidia-curand-cu11==10.3.0.86 (from torch)

Downloading https://download.pytorch.org/whl/cu118/nvidia_curand_cu11-10.3.0.86-py3-none-manylinux1_x86_64.whl (58.1 MB)

58.1/58.1 MB 55.9 MB/s eta 0:00:000:0100:01

```

Collecting nvidia-cusolver-cu11==11.4.1.48 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_cusolver_cu11-11.4.1.48-py3-none-manylinux1_x86_64.whl (128.2 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 128.2/128.2 MB 29.7 MB/s eta 0:00:000:0100:01
Collecting nvidia-cuspars-cu11==11.7.5.86 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_cuspars-cu11-11.7.5.86-py3-none-manylinux1_x86_64.whl (204.1 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 204.1/204.1 MB 20.0 MB/s eta 0:00:000:0100:01
Collecting nvidia-nccl-cu11==2.21.5 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_nccl_cu11-2.21.5-py3-none-manylinux2014_x86_64.whl (147.8 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 147.8/147.8 MB 27.6 MB/s eta 0:00:000:0100:01
Collecting nvidia-nvtx-cu11==11.8.86 (from torch)
  Downloading https://download.pytorch.org/whl/cu118/nvidia_nvtx_cu11-11.8.86-py3-none-manylinux1_x86_64.whl (99 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 99.1/99.1 kB 25.3 MB/s eta 0:00:00
Collecting triton==3.3.1 (from torch)
  Downloading https://download-r2.pytorch.org/whl/triton-3.3.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.5 kB)
Requirement already satisfied: setuptools>=40.8.0 in /usr/local/lib/python3.11/dist-packages (from triton==3.3.1->torch) (69.0.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from torchvision) (1.26.4)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.11/dist-packages (from torchvision) (9.5.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch) (2.1.4)
Downloading https://download.pytorch.org/whl/cu118/torch-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl (905.3 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 905.3/905.3 MB 3.7 MB/s eta 0:00:00:00:0100:02
Downloading https://download-r2.pytorch.org/whl/triton-3.3.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (155.7 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 155.7/155.7 MB 16.1 MB/s eta 0:00:000:0100:01
Downloading https://download-r2.pytorch.org/whl/cu118/torchvision-0.22.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl (6.7 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.7/6.7 MB 92.1 MB/s eta 0:00:00:00:0100:01
Downloading https://download-r2.pytorch.org/whl/cu118/torchaudio-2.7.1%2Bcu118-cp311-cp311-manylinux_2_28_x86_64.whl (3.3 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.3/3.3 MB 49.8 MB/s eta 0:00:00:00:01
Installing collected packages: triton, nvidia-nvtx-cu11, nvidia-nccl-cu11, nvidia-cuspars-cu11, nvidia-curand-cu11, nvidia-cufft-cu11, nvidia-cuda-runtime-cu11, nvidia-cuda-nvrtc-cu11, nvidia-cuda-cupti-cu11, nvidia-cublas-cu11, nvidia-cusolver-cu11, nvidia-cudnn-cu11, torch, torchvision, torchaudio
Attempting uninstall: triton
  Found existing installation: triton 3.6.0
  Uninstalling triton-3.6.0:
    Successfully uninstalled triton-3.6.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
sentence-transformers 2.2.2 requires transformers<5.0.0,>=4.6.0, but you have transformers 5.4.0 which is incompatible.
Successfully installed nvidia-cublas-cu11-11.11.3.6 nvidia-cuda-cupti-cu11-11.8.87 nvidia-cuda-nvrtc-cu11-11.8.89 nvidia-cuda-runtime-cu11-11.8.89 nvidia-cudnn-cu11-9.1.0.70 nvidia-

```

cufft-cu11-10.9.0.58 nvidia-curand-cu11-10.3.0.86 nvidia-cusolver-cu11-11.4.1.48 nvidia-cusparse-cu11-11.7.5.86 nvidia-nccl-cu11-2.21.5 nvidia-nvtx-cu11-11.8.86 torch-2.7.1+cu118 torchaudio-2.7.1+cu118 torchvision-0.22.1+cu118 triton-3.3.1

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

Note: you may need to restart the kernel to use updated packages.

```
In [7]: # JupyterLab / fresh GPU machine setup
import sys
print(sys.executable)
```

```
Out[7]: /usr/local/bin/python3
```

Import Required Libraries

Set Random Seed for Reproducibility

```
In [1]: import os
import re
import random
import numpy as np
import pandas as pd
import torch

from tqdm import tqdm
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix, classification_report
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)
```

Upload Dataset

Upload **sample_full.csv** from the Monash dataset.

The dataset contains:

- Learning_outcome
- Bloom taxonomy one-hot columns

```
In [2]: import os
import pandas as pd

CSV_PATH = "content/sample_full.csv" # apni CSV ka exact naam do
assert os.path.exists(CSV_PATH), f"CSV file not found: {CSV_PATH}"

df = pd.read_csv(CSV_PATH)
print("Dataset size:", len(df))
df.head()
```

Out[2]: Dataset size: 21380

Out[2]:

	Learning_outcome	Remember	Understand	Apply	Analyze	Evaluate	Create
0	Analyze the health economic implications of e...	NaN	NaN	NaN	1.0	NaN	NaN
1	Apply research skills to operate effectively ...	NaN	NaN	1.0	NaN	NaN	NaN
2	Assess and synthesise diverse information abo...	NaN	NaN	NaN	NaN	1.0	1.0
3	Describe the general characteristics of the m...	NaN	1.0	NaN	NaN	NaN	NaN
4	Evaluate the different models of perioperativ...	NaN	NaN	NaN	NaN	1.0	NaN

Convert One-Hot Encoding to Single Label

Bloom taxonomy columns:

- Remember
- Understand
- Apply
- Analyze
- Evaluate
- Create

```
In [3]: LABELS = ["Remember", "Understand", "Apply", "Analyze", "Evaluate", "Create"]

required_cols = ["Learning_outcome"] + LABELS
missing_cols = [c for c in required_cols if c not in df.columns]
assert not missing_cols, f"Missing columns: {missing_cols}"

df["label"] = df[LABELS].idxmax(axis=1)
df = df[["Learning_outcome", "label"]].dropna().reset_index(drop=True)

print(df.head())
print(df["label"].value_counts())
```

	Learning_outcome	label
0	Analyze the health economic implications of e...	Analyze
1	Apply research skills to operate effectively ...	Apply
2	Assess and synthesise diverse information abo...	Evaluate
3	Describe the general characteristics of the m...	Understand
4	Evaluate the different models of perioperativ...	Evaluate

label

Apply	5801
Understand	5694
Analyze	3096
Create	2955
Evaluate	2649
Remember	1185

Name: count, dtype: int64

Train / Test Split

80% training data

20% testing data

```
In [4]: train_df, test_df = train_test_split(
        df,
        test_size=0.2,
        random_state=SEED,
        stratify=df["label"]
    )

print("Train size:", len(train_df))
print("Test size:", len(test_df))
```

```
Out[4]: Train size: 17104
        Test size: 4276
```

Load Model (4-bit Quantization)

4-bit quantization allows large models to run on Colab T4 GPU.

```
In [5]: import torch
import platform

print("Python:", platform.python_version())
print("PyTorch:", torch.__version__)
print("CUDA available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
    print("CUDA device count:", torch.cuda.device_count())
    print("Current device:", torch.cuda.current_device())
```

```
else:
    print("Running on CPU")
```

Out[5]:

```
Python: 3.11.7
PyTorch: 2.7.1+cu118
CUDA available: True
GPU: NVIDIA RTX A6000
CUDA device count: 1
Current device: 0
```

In [6]:

```
import torch
if torch.cuda.is_available():
    torch.cuda.empty_cache()
    print("CUDA cache cleared.")
else:
    print("CUDA is not available, no cache to clear.")
```

Out[6]:

```
CUDA cache cleared.
```

In [7]:

```
from huggingface_hub import login

HF_TOKEN = "hf_byvpkLIEiQdmtaXnvSRdYsebl1NHbXtnPs"

login(HF_TOKEN)
```

In [8]:

```
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig

MODEL_REGISTRY = {
    # Already using
    "mistral_7b_instruct_v02": "mistralai/Mistral-7B-Instruct-v0.3",
    "llama3_8b_instruct": "meta-llama/Llama-3.1-8B-Instruct",
    "phi35_mini_4k_instruct": "microsoft/Phi-3.5-mini-instruct",
    "qwen2_7b_instruct": "Qwen/Qwen2-7B-Instruct",

    # Recommended additions
    "llama31_8b_instruct": "meta-llama/Llama-3.1-8B-Instruct",
    "qwen25_7b_instruct": "Qwen/Qwen2.5-7B-Instruct",

    # Smaller / faster options
    "qwen25_3b_instruct": "Qwen/Qwen2.5-3B-Instruct",
    "qwen25_1_5b_instruct": "Qwen/Qwen2.5-1.5B-Instruct",
    "smollm2_1_7b_instruct": "HuggingFaceTB/SmolLM2-1.7B-Instruct",
    "gemma2_2b_instruct": "google/gemma-2-2b-it",
    "gemma2_9b_instruct": "google/gemma-2-9b-it",

    # Good experimental middle tier
    "phi35_mini_instruct": "microsoft/Phi-4-mini-instruct",
    "mistral_nemo_instruct": "mistralai/Mistral-Nemo-Instruct-2407"
}
```

In [9]:

```
def load_model_and_tokenizer(model_id):

    bnb_config = BitsAndBytesConfig(
        load_in_4bit=True,
        bnb_4bit_quant_type="nf4",
        bnb_4bit_use_double_quant=True,
        bnb_4bit_compute_dtype=torch.float16
    )

    tokenizer = AutoTokenizer.from_pretrained(model_id, use_fast=True)

    # PAD token fix
    if tokenizer.pad_token is None:
        tokenizer.pad_token = tokenizer.eos_token

    model = AutoModelForCausalLM.from_pretrained(
        model_id,
        quantization_config=bnb_config,
        device_map="auto",
        torch_dtype=torch.float16
    )

    model.config.pad_token_id = tokenizer.pad_token_id

    return tokenizer, model
```

Prompt Design

We use a **zero-shot classification prompt**.

The model must return only one Bloom taxonomy label.

In [10]:

```
def create_prompt(text):
    return f"""You are an expert in Bloom's Taxonomy classification.

    Classify the following learning outcome into exactly ONE of these labels:

    Remember
    Understand
    Apply
    Analyze
    Evaluate
    Create

    Rules:
    - Return ONLY one label.
    - Do not explain.
    - Do not repeat the list.
    - Do not write any extra text.

    Learning Outcome:
    {text}
```

```
"""
```

Normalize Model Output

```
In [11]: LABELS = ["Remember", "Understand", "Apply", "Analyze", "Evaluate", "Create"]

def normalize_output(text):
    text = str(text).strip().lower()

    matches = []

    for label in LABELS:
        pattern = r"\b" + re.escape(label.lower()) + r"\b"
        if re.search(pattern, text):
            matches.append(label)

    if matches:
        return matches[-1]

    return "Invalid"
```

LLM Prediction Function

```
In [12]: def predict(text, tokenizer, model, max_new_tokens=12):
    prompt = create_prompt(text)

    inputs = tokenizer(
        prompt,
        return_tensors="pt",
        truncation=True,
        max_length=512
    ).to(model.device)

    input_length = inputs["input_ids"].shape[1]

    with torch.no_grad():
        output = model.generate(
            **inputs,
            max_new_tokens=max_new_tokens,
            do_sample=False,
            pad_token_id=tokenizer.pad_token_id
        )

    generated_tokens = output[0][input_length:]
    decoded = tokenizer.decode(generated_tokens, skip_special_tokens=True).strip()

    return decoded
```

Evaluate Model on Test Data

In [13]:

```
import gc
import torch

def clear_memory():
    gc.collect()
    if torch.cuda.is_available():
        torch.cuda.empty_cache()
```

In [14]:

```
sample_model_name = list(MODEL_REGISTRY.keys())[0]
sample_model_id = MODEL_REGISTRY[sample_model_name]

print("Testing model:", sample_model_name)

tokenizer, model = load_model_and_tokenizer(sample_model_id)

for i in range(10):
    text = test_df.iloc[i]["Learning_outcome"]
    raw_pred = predict(text, tokenizer, model)
    clean_pred = normalize_output(raw_pred)

    print("=" * 80)
    print("TEXT:", text)
    print("RAW OUTPUT:", raw_pred)
    print("NORMALIZED:", clean_pred)

del model
del tokenizer
clear_memory()
```

Out[14]: Testing model: mistral_7b_instruct_v02

Out[14]: config.json: 0%| | 0.00/601 [00:00<?, ?B/s]

Out[14]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[14]: tokenizer.json: 0.00B [00:00, ?B/s]

Out[14]: tokenizer.model: 0%| | 0.00/587k [00:00<?, ?B/s]

Out[14]: special_tokens_map.json: 0%| | 0.00/414 [00:00<?, ?B/s]

Out[14]: `torch\_dtype` is deprecated! Use `dtype` instead!

Out[14]: model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[14]: Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[14]: Fetching 3 files: 0%| | 0/3 [00:00<?, ?it/s]

Out[14]: Loading weights: 0%| | 0/291 [00:00<?, ?it/s]

Out[14]: generation_config.json: 0%| | 0.00/116 [00:00<?, ?B/s]

Out[14]: =====
TEXT: appreciate science and technologies as a way of thinking and acting that generates knowledge and which represents cultures and societies
RAW OUTPUT: Analyze
NORMALIZED: Analyze
=====

TEXT: Critically reflect on your own design practice and projects to date;
RAW OUTPUT: identify areas for improvement and develop a plan to address these
NORMALIZED: Invalid
=====

TEXT: Reflect on the impact of recent developments in science and philosophy on our most fundamental views about ourselves and our place in the world;
RAW OUTPUT: Analyze
NORMALIZED: Analyze
=====

TEXT: Integrate and summarise information obtained during psychological assessments in the form of a professional report;
RAW OUTPUT: Analyze
NORMALIZED: Analyze
=====

TEXT: Have developed the ability to apply philosophical theories of justice in conflict to actual cases from recent history;
RAW OUTPUT: Apply
NORMALIZED: Apply
=====

TEXT: Describe medical and professional ethics and different ethical perspectives as related to healthcare
RAW OUTPUT: Analyze
NORMALIZED: Analyze
=====

TEXT: Conduct research in topics covered in Current Issues in Workplace Law based on knowledge of appropriate research principles and methods
RAW OUTPUT: Analyze
NORMALIZED: Analyze
=====

TEXT: Describe and implement patient care and quality assurance principles to sonographic imaging.
RAW OUTPUT: "Apply"
NORMALIZED: Apply
=====

TEXT: Demonstrate the ability to research and write from an informed position about the body in contemporary art.
RAW OUTPUT: "Analyze"

NORMALIZED: Analyze

TEXT: Identify data resources and standards.

RAW OUTPUT: "Understand"

NORMALIZED: Understand

```
In [15]: from sklearn.metrics import (
    accuracy_score,
    f1_score,
    precision_score,
    recall_score,
    confusion_matrix,
    classification_report,
    balanced_accuracy_score,
    matthews_corrcoef,
    cohen_kappa_score
)

all_results = []
all_predictions = {}

for model_name, model_id in MODEL_REGISTRY.items():
    print("=" * 80)
    print(f"Loading model: {model_name} -> {model_id}")

    if torch.cuda.is_available():
        torch.cuda.empty_cache()

    tokenizer, model = None, None

    try:
        tokenizer, model = load_model_and_tokenizer(model_id)

        predictions = []
        for _, row in tqdm(test_df.iterrows(), total=len(test_df), desc=model_name):
            pred = predict(row["Learning_outcome"], tokenizer, model)
            predictions.append(pred)

        result_df = test_df.copy()
        result_df["raw_prediction"] = predictions
        result_df["prediction"] = result_df["raw_prediction"].apply(normalize_output)

        result_df = result_df.dropna(subset=["label", "prediction"])
        y_true = result_df["label"].astype(str).str.strip()
        y_pred = result_df["prediction"].astype(str).str.strip()

        accuracy = accuracy_score(y_true, y_pred)
        macro_f1 = f1_score(y_true, y_pred, average="macro", zero_division=0)
        micro_f1 = f1_score(y_true, y_pred, average="micro", zero_division=0)
        weighted_f1 = f1_score(y_true, y_pred, average="weighted", zero_division=0)
        macro_precision = precision_score(y_true, y_pred, average="macro", zero_division=0)
        macro_recall = recall_score(y_true, y_pred, average="macro", zero_division=0)
        balanced_acc = balanced_accuracy_score(y_true, y_pred)
        mcc = matthews_corrcoef(y_true, y_pred)
```

```

kappa = cohen_kappa_score(y_true, y_pred)
cm = confusion_matrix(y_true, y_pred)

print("Accuracy:", accuracy)
print("Macro F1:", macro_f1)
print("Micro F1:", micro_f1)
print("Weighted F1:", weighted_f1)
print("Macro Precision:", macro_precision)
print("Macro Recall:", macro_recall)
print("Balanced Accuracy:", balanced_acc)
print("Matthews Corrcoef:", mcc)
print("Cohen Kappa:", kappa)
print("\nClassification Report:")
print(classification_report(y_true, y_pred, zero_division=0))
print("\nConfusion Matrix:")
print(cm)

result_df.to_csv(f"{model_name}_predictions.csv", index=False)

all_results.append({
    "model_name": model_name,
    "model_id": model_id,
    "accuracy": accuracy,
    "macro_f1": macro_f1,
    "micro_f1": micro_f1,
    "weighted_f1": weighted_f1,
    "macro_precision": macro_precision,
    "macro_recall": macro_recall,
    "balanced_accuracy": balanced_acc,
    "mcc": mcc,
    "cohen_kappa": kappa
})

all_predictions[model_name] = result_df

except Exception as e:
    print(f"Error while testing {model_name}: {e}")
    all_results.append({
        "model_name": model_name,
        "model_id": model_id,
        "accuracy": None,
        "macro_f1": None,
        "micro_f1": None,
        "weighted_f1": None,
        "macro_precision": None,
        "macro_recall": None,
        "balanced_accuracy": None,
        "mcc": None,
        "cohen_kappa": None
    })

finally:
    if model is not None:
        del model
    if tokenizer is not None:

```

```
del tokenizer
clear_memory()
```

```
Out[15]:
=====
Loading model: mistral_7b_instruct_v02 -> mistralai/Mistral-7B-Instruct-v0.3
```

```
Out[15]: Loading weights:  0%|          | 0/291 [00:00<?, ?it/s]
```

```
Out[15]: mistral_7b_instruct_v02: 100%|██████████| 4276/4276 [15:32<00:00,  4.59it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
  warnings.warn("y_pred contains classes not in y_true")
```

```
Out[15]: Accuracy: 0.43662301216089805
Macro F1: 0.35214507631115965
Micro F1: 0.43662301216089805
Weighted F1: 0.45988834556921315
Macro Precision: 0.49606202345358136
Macro Recall: 0.35813610258614376
Balanced Accuracy: 0.4178254530171677
Matthews Corrcoeff: 0.3748485478226825
Cohen Kappa: 0.3230246058123273
```

Classification Report:

	precision	recall	f1-score	support
Analyze	0.23	0.87	0.36	619
Apply	0.80	0.44	0.56	1160
Create	0.66	0.35	0.46	591
Evaluate	0.60	0.50	0.55	530
Invalid	0.00	0.00	0.00	0
Remember	0.30	0.05	0.09	237
Understand	0.89	0.30	0.45	1139
accuracy			0.44	4276
macro avg	0.50	0.36	0.35	4276
weighted avg	0.67	0.44	0.46	4276

Confusion Matrix:

```
[[536  7  8 40 28  0  0]
 [478 505 69 60 19 14 15]
 [269 59 206 31 18  1  7]
 [225  9  15 267 12  1  1]
 [ 0  0  0  0  0  0  0]
 [182 11  3  6  1 13 21]
 [684 38 13 41  9 14 340]]
```

```
=====
Loading model: llama3_8b_instruct -> meta-llama/Llama-3.1-8B-Instruct
```

```
Out[15]:
```

```

Out[15]: config.json:  0%|          | 0.00/855 [00:00<?, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]: special_tokens_map.json:  0%|          | 0.00/296 [00:00<?, ?B/s]

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]: Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]: Fetching 4 files:  0%|          | 0/4 [00:00<?, ?it/s]

Out[15]: Loading weights:  0%|          | 0/291 [00:00<?, ?it/s]

Out[15]: generation_config.json:  0%|          | 0.00/184 [00:00<?, ?B/s]

llama3_8b_instruct:  0%|          | 0/4276 [00:00<?, ?it/s]The following generation flags
are not valid and may be ignored: ['temperature', 'top_p']. Set `TRANSFORMERS_VERBOSITY=info`
for more details.
llama3_8b_instruct: 100%|██████████| 4276/4276 [39:53<00:00, 1.79it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
  warnings.warn("y_pred contains classes not in y_true")

Out[15]:

Accuracy: 0.17563143124415342
Macro F1: 0.15300160001073476
Micro F1: 0.17563143124415342
Weighted F1: 0.20132458636526207
Macro Precision: 0.2837474574777941
Macro Recall: 0.13145056309996697
Balanced Accuracy: 0.1533589902832948
Matthews Corrcoeff: 0.08160598227871498
Cohen Kappa: 0.06943782138965038

Classification Report:

```

	precision	recall	f1-score	support
Analyze	0.17	0.19	0.18	619
Apply	0.36	0.34	0.35	1160
Create	0.35	0.13	0.19	591
Evaluate	0.33	0.21	0.25	530
Invalid	0.00	0.00	0.00	0
Remember	0.10	0.01	0.02	237
Understand	0.68	0.04	0.08	1139

accuracy			0.18	4276
macro avg	0.28	0.13	0.15	4276
weighted avg	0.40	0.18	0.20	4276

Confusion Matrix:

```
[[119 130 33 79 255 1 2]
 [142 396 55 47 506 4 10]
 [ 66 154 77 12 276 2 4]
 [ 94 81 16 109 228 0 2]
 [ 0 0 0 0 0 0 0]
 [ 35 81 4 6 104 2 5]
 [241 248 34 78 478 12 48]]
```

=====

Loading model: phi35_mini_4k_instruct -> microsoft/Phi-3.5-mini-instruct

config.json: 0.00B [00:00, ?B/s]

This model config has set a `rope\_parameters['original\_max\_position\_embeddings']` field, to be used together with `max\_position\_embeddings` to determine a scaling factor. Please set the `factor` field of `rope\_parameters` with this ratio instead -- we recommend the use of this field over `original\_max\_position\_embeddings`, as it is compatible with most model architectures.

tokenizer_config.json: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

tokenizer.model: 0%| | 0.00/500k [00:00<?, ?B/s]

added_tokens.json: 0%| | 0.00/306 [00:00<?, ?B/s]

special_tokens_map.json: 0%| | 0.00/665 [00:00<?, ?B/s]

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Loading weights: 0%| | 0/195 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/195 [00:00<?, ?B/s]

phi35_mini_4k_instruct: 100%|██████████| 4276/4276 [23:04<00:00, 3.09it/s]

y_pred contains classes not in y_true

warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.6038353601496725
Macro F1: 0.4576890348752175
Micro F1: 0.6038353601496725
Weighted F1: 0.6016796747346466
Macro Precision: 0.4936578482580489
Macro Recall: 0.4660687343201178
Balanced Accuracy: 0.5437468567068041
Matthews Corrcoef: 0.5113458750324916
Cohen Kappa: 0.5058659262465972

Classification Report:

	precision	recall	f1-score	support
Analyze	0.43	0.68	0.53	619
Apply	0.66	0.64	0.65	1160
Create	0.74	0.48	0.58	591
Evaluate	0.49	0.72	0.58	530
Invalid	0.00	0.00	0.00	0
Remember	0.35	0.10	0.16	237
Understand	0.78	0.63	0.70	1139
accuracy			0.60	4276
macro avg	0.49	0.47	0.46	4276
weighted avg	0.63	0.60	0.60	4276

Confusion Matrix:

```
[[424  51   8 122   1   0  13]
 [147 748  69 100   0  19  77]
 [ 49 188 284  44   4   2  20]
 [ 85  33  11 380   1   0  20]
 [  0   0   0   0   0   0   0]
 [ 76  36   1  24   4  24  72]
 [206  71  12 105   0  23 722]]
```

Loading model: qwen2_7b_instruct -> Qwen/Qwen2-7B-Instruct

config.json: 0%| | 0.00/663 [00:00<?, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

vocab.json: 0.00B [00:00, ?B/s]

merges.txt: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]:

Fetching 4 files: 0%| | 0/4 [00:00<?, ?it/s]

Out[15]:

Loading weights: 0%| | 0/339 [00:00<?, ?it/s]

Out[15]:

generation_config.json: 0%| | 0.00/243 [00:00<?, ?B/s]

Out[15]:

qwen2_7b_instruct: 0%| | 0/4276 [00:00<?, ?it/s]The following generation flags are not valid and may be ignored: ['temperature', 'top_p', 'top_k']. Set `TRANSFORMERS\_VERBOSITY=info` for more details.
qwen2_7b_instruct: 100%|██████████| 4276/4276 [15:31<00:00, 4.59it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning: y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Out[15]:

Accuracy: 0.5318054256314313
Macro F1: 0.43046306756480435
Micro F1: 0.5318054256314313
Weighted F1: 0.5456156031191579
Macro Precision: 0.5038186769505585
Macro Recall: 0.4361460459284344
Balanced Accuracy: 0.5088370535831735
Matthews Corrcoeff: 0.4469308139800939
Cohen Kappa: 0.4313677573228766

Classification Report:

	precision	recall	f1-score	support
Analyze	0.29	0.61	0.39	619
Apply	0.72	0.61	0.66	1160
Create	0.57	0.66	0.61	591
Evaluate	0.55	0.65	0.60	530
Invalid	0.00	0.00	0.00	0
Remember	0.49	0.15	0.23	237
Understand	0.90	0.36	0.52	1139
accuracy			0.53	4276
macro avg	0.50	0.44	0.43	4276
weighted avg	0.65	0.53	0.55	4276

Confusion Matrix:

```
[[379  43  35 115  45   1   1]
 [141 709 195  54  15  18  28]
 [ 62  86 393  30  16   1   3]
 [127  11  41 345   6   0   0]
 [  0   0   0   0   0   0   0]
```

```
[113  50   5  17   3  36  13]
[495  81  23  63  48  17 412]]
```

```
=====
Loading model: llama31_8b_instruct -> meta-llama/Llama-3.1-8B-Instruct
```

Out[15]:

```
Loading weights:   0%|          | 0/291 [00:00<?, ?it/s]
```

Out[15]:

```
llama31_8b_instruct: 100%|██████████| 4276/4276 [39:52<00:00,  1.79it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
  warnings.warn("y_pred contains classes not in y_true")
```

Out[15]:

```
Accuracy: 0.17563143124415342
Macro F1: 0.15300160001073476
Micro F1: 0.17563143124415342
Weighted F1: 0.20132458636526207
Macro Precision: 0.2837474574777941
Macro Recall: 0.13145056309996697
Balanced Accuracy: 0.1533589902832948
Matthews Corrcoeff: 0.08160598227871498
Cohen Kappa: 0.06943782138965038
```

Classification Report:

	precision	recall	f1-score	support
Analyze	0.17	0.19	0.18	619
Apply	0.36	0.34	0.35	1160
Create	0.35	0.13	0.19	591
Evaluate	0.33	0.21	0.25	530
Invalid	0.00	0.00	0.00	0
Remember	0.10	0.01	0.02	237
Understand	0.68	0.04	0.08	1139
accuracy			0.18	4276
macro avg	0.28	0.13	0.15	4276
weighted avg	0.40	0.18	0.20	4276

Confusion Matrix:

```
[[119 130  33  79 255   1   2]
 [142 396  55  47 506   4  10]
 [ 66 154  77  12 276   2   4]
 [ 94  81  16 109 228   0   2]
 [  0   0   0   0   0   0   0]
 [ 35  81   4   6 104   2   5]
 [241 248  34  78 478  12  48]]
```

```
=====
Loading model: qwen25_7b_instruct -> Qwen/Qwen2.5-7B-Instruct
```

Out[15]:

```
config.json:   0%|          | 0.00/663 [00:00<?, ?B/s]
```

Out[15]:

tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]:

vocab.json: 0.00B [00:00, ?B/s]

Out[15]:

merges.txt: 0.00B [00:00, ?B/s]

Out[15]:

tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]:

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]:

Fetching 4 files: 0%| | 0/4 [00:00<?, ?it/s]

Out[15]:

Loading weights: 0%| | 0/339 [00:00<?, ?it/s]

Out[15]:

generation_config.json: 0%| | 0.00/243 [00:00<?, ?B/s]

Out[15]:

qwen25_7b_instruct: 100%|██████████| 4276/4276 [17:53<00:00, 3.98it/s]

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning: y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Out[15]:

Accuracy: 0.6115528531337698
Macro F1: 0.48290399269906636
Micro F1: 0.6115528531337698
Weighted F1: 0.6128618062477388
Macro Precision: 0.49774474963533777
Macro Recall: 0.4883411193256757
Balanced Accuracy: 0.569731305879955
Matthews Corrccoef: 0.5200638579057925
Cohen Kappa: 0.5167511449750103

Classification Report:

	precision	recall	f1-score	support
Analyze	0.58	0.43	0.49	619
Apply	0.70	0.66	0.68	1160
Create	0.52	0.64	0.57	591
Evaluate	0.45	0.72	0.55	530
Invalid	0.00	0.00	0.00	0
Remember	0.47	0.31	0.38	237
Understand	0.77	0.66	0.71	1139
accuracy			0.61	4276
macro avg	0.50	0.49	0.48	4276
weighted avg	0.63	0.61	0.61	4276

Confusion Matrix:

```
[[265  70  78 179   1   2  24]
 [ 42 761 126 105   0  28  98]
 [ 16 110 379  41   1   7  37]
 [ 45  26  54 380   0   2  23]
 [  0   0   0   0   0   0   0]
 [ 23  44  15  31   1  74  49]
 [ 63  74  82 117   3  44 756]]
```

=====

Loading model: qwen25_3b_instruct -> Qwen/Qwen2.5-3B-Instruct

Out[15]:

config.json: 0%| | 0.00/661 [00:00<?, ?B/s]

Out[15]:

tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]:

vocab.json: 0.00B [00:00, ?B/s]

Out[15]:

merges.txt: 0.00B [00:00, ?B/s]

Out[15]:

tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]:

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]:

Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Out[15]:

Loading weights: 0%| | 0/434 [00:00<?, ?it/s]

Out[15]:

generation_config.json: 0%| | 0.00/242 [00:00<?, ?B/s]

Out[15]:

qwen25_3b_instruct: 100%|██████████| 4276/4276 [16:45<00:00, 4.25it/s]

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Out[15]:

Accuracy: 0.32343311506080447
Macro F1: 0.22974408798124127
Micro F1: 0.32343311506080447
Weighted F1: 0.2902284217060754
Macro Precision: 0.3884986139504725
Macro Recall: 0.285230929768853
Balanced Accuracy: 0.33276941806366184

Matthews Corrcoef: 0.23091998671711905
Cohen Kappa: 0.2019870995160561

Classification Report:

	precision	recall	f1-score	support
Analyze	0.26	0.77	0.39	619
Apply	0.70	0.34	0.46	1160
Create	0.48	0.28	0.35	591
Evaluate	0.22	0.55	0.31	530
Invalid	0.00	0.00	0.00	0
Remember	0.20	0.01	0.02	237
Understand	0.87	0.04	0.08	1139
accuracy			0.32	4276
macro avg	0.39	0.29	0.23	4276
weighted avg	0.56	0.32	0.29	4276

Confusion Matrix:

```
[[478 15 11 107 8 0 0]
 [301 400 107 310 32 6 4]
 [105 49 167 254 14 1 1]
 [199 13 13 291 13 0 1]
 [ 0 0 0 0 0 0 0]
 [125 30 2 69 8 2 1]
 [644 63 51 301 34 1 45]]
```

=====

Out[15]: Loading model: qwen25_1_5b_instruct -> Qwen/Qwen2.5-1.5B-Instruct

Out[15]: config.json: 0%| | 0.00/660 [00:00<?, ?B/s]

Out[15]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: vocab.json: 0.00B [00:00, ?B/s]

Out[15]: merges.txt: 0.00B [00:00, ?B/s]

Out[15]: tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]: model.safetensors: 0%| | 0.00/3.09G [00:00<?, ?B/s]

Out[15]: Loading weights: 0%| | 0/338 [00:00<?, ?it/s]

Out[15]: generation_config.json: 0%| | 0.00/242 [00:00<?, ?B/s]

y_pred contains classes not in y_true

Out[15]: warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.4567352666043031

Macro F1: 0.30624009729233104

Micro F1: 0.4567352666043031

Weighted F1: 0.4395190672151548

Macro Precision: 0.3599435066745224

Macro Recall: 0.317043931717797

Balanced Accuracy: 0.3698845870040965

Matthews Corrcoeff: 0.3250709318069589

Cohen Kappa: 0.31704721826537674

Classification Report:

	precision	recall	f1-score	support
Analyze	0.37	0.71	0.49	619
Apply	0.56	0.62	0.59	1160
Create	0.35	0.16	0.22	591
Evaluate	0.71	0.22	0.33	530
Invalid	0.00	0.00	0.00	0
Remember	0.00	0.00	0.00	237
Understand	0.52	0.51	0.51	1139
accuracy			0.46	4276
macro avg	0.36	0.32	0.31	4276
weighted avg	0.48	0.46	0.44	4276

Confusion Matrix:

```
[[441  52  20  11  33   1  61]
 [115 725  54  13  61   0 192]
 [100 250  92   4  43   0 102]
 [254  66  19 115  24   0  52]
 [  0   0   0   0   0   0   0]
 [ 42  33   6   5  19   0 132]
 [236 158  71  13  81   0 580]]
```

=====

Out[15]: Loading model: smollm2_1_7b_instruct -> HuggingFaceTB/SmolLM2-1.7B-Instruct

Out[15]: config.json: 0%| | 0.00/908 [00:00<?, ?B/s]

Out[15]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: vocab.json: 0.00B [00:00, ?B/s]

Out[15]: merges.txt: 0.00B [00:00, ?B/s]

Out[15]: special_tokens_map.json: 0%| | 0.00/655 [00:00<?, ?B/s]

```
Out[15]: tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]: model.safetensors:  0%|          | 0.00/3.42G [00:00<?, ?B/s]

Out[15]: Loading weights:  0%|          | 0/218 [00:00<?, ?it/s]

Out[15]: generation_config.json:  0%|          | 0.00/132 [00:00<?, ?B/s]

smollm2_1_7b_instruct: 100%|██████████| 4276/4276 [10:23<00:00, 6.86it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
Out[15]: warnings.warn("y_pred contains classes not in y_true")
```

Accuracy: 0.022918615528531337
Macro F1: 0.02484740274114969
Micro F1: 0.022918615528531337
Weighted F1: 0.01592309719089855
Macro Precision: 0.23860355665432728
Macro Recall: 0.05084911014741377
Balanced Accuracy: 0.0593239618386494
Matthews Corrcoeff: 0.014031597926021358
Cohen Kappa: 0.007783222542603885

Classification Report:

	precision	recall	f1-score	support
Analyze	0.00	0.00	0.00	619
Apply	0.59	0.01	0.02	1160
Create	0.62	0.01	0.02	591
Evaluate	0.09	0.00	0.01	530
Invalid	0.00	0.00	0.00	0
Remember	0.08	0.33	0.13	237
Understand	0.29	0.00	0.00	1139
accuracy			0.02	4276
macro avg	0.24	0.05	0.02	4276
weighted avg	0.34	0.02	0.02	4276

Confusion Matrix:

```
[ [ 0  1  0  2 475 141  0]
  [ 1 10  1  4 959 183  2]
  [ 2  4  5  5 434 140  1]
  [ 0  0  0  2 390 138  0]
  [ 0  0  0  0  0  0  0]
  [ 0  0  0  2 154  79  2]
  [ 0  2  2  7 824 302  2]]
```

=====

```
Out[15]: Loading model: gemma2_2b_instruct -> google/gemma-2-2b-it
```

Out[15]: config.json: 0%| | 0.00/838 [00:00<?, ?B/s]

Out[15]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: tokenizer.json: 0%| | 0.00/17.5M [00:00<?, ?B/s]

Out[15]: special_tokens_map.json: 0%| | 0.00/636 [00:00<?, ?B/s]

Out[15]: model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]: Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]: Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Out[15]: Loading weights: 0%| | 0/288 [00:00<?, ?it/s]

Out[15]: generation_config.json: 0%| | 0.00/187 [00:00<?, ?B/s]

gemma2_2b_instruct: 100%|██████████| 4276/4276 [30:54<00:00, 2.31it/s]

Out[15]: /usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning: y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.23971000935453696
Macro F1: 0.24128383896332298
Micro F1: 0.23971000935453696
Weighted F1: 0.3131329170599725
Macro Precision: 0.41849639933129995
Macro Recall: 0.18743944971242138
Balanced Accuracy: 0.21867935799782492
Matthews Corrcoef: 0.19395163040253038
Cohen Kappa: 0.15845284072675636

Classification Report:

	precision	recall	f1-score	support
Analyze	0.38	0.32	0.35	619
Apply	0.52	0.30	0.38	1160
Create	0.54	0.24	0.33	591
Evaluate	0.44	0.26	0.33	530
Invalid	0.00	0.00	0.00	0
Remember	0.40	0.02	0.03	237
Understand	0.66	0.17	0.27	1139
accuracy			0.24	4276
macro avg	0.42	0.19	0.24	4276
weighted avg	0.52	0.24	0.31	4276

Confusion Matrix:

```
[[200  46  10  39 312   0  12]
 [ 79 349  82  56 555   3  36]
 [ 37 114 142  31 253   1  13]
 [ 48  40  18 140 276   1   7]
 [  0   0   0   0   0   0   0]
 [ 34  34   3  18 113   4  31]
 [134  89   9  35 681   1 190]]
```

Out[15]: =====
Loading model: gemma2_9b_instruct -> google/gemma-2-9b-it

Out[15]: config.json: 0%| | 0.00/857 [00:00<?, ?B/s]

Out[15]: tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]: tokenizer.json: 0%| | 0.00/17.5M [00:00<?, ?B/s]

Out[15]: special_tokens_map.json: 0%| | 0.00/636 [00:00<?, ?B/s]

Out[15]: model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]: Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]: Fetching 4 files: 0%| | 0/4 [00:00<?, ?it/s]

Out[15]: Loading weights: 0%| | 0/464 [00:00<?, ?it/s]

Out[15]: generation_config.json: 0%| | 0.00/173 [00:00<?, ?B/s]

gemma2_9b_instruct: 100%|██████████| 4276/4276 [16:09<00:00, 4.41it/s]

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:

Out[15]: y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.04373246024321796

Macro F1: 0.056122951839338776

Micro F1: 0.04373246024321796

Weighted F1: 0.07862798766717917

Macro Precision: 0.419988859342994

Macro Recall: 0.03123357633531686

Balanced Accuracy: 0.036439172391203

Matthews Corrcoeff: 0.08830271771298312

Cohen Kappa: 0.029588977809529182

Classification Report:

	precision	recall	f1-score	support
Analyze	0.57	0.08	0.14	619
Apply	0.75	0.09	0.16	1160
Create	0.53	0.03	0.06	591
Evaluate	0.09	0.01	0.01	530
Invalid	0.00	0.00	0.00	0
Remember	0.00	0.00	0.00	237
Understand	1.00	0.01	0.02	1139
accuracy			0.04	4276
macro avg	0.42	0.03	0.06	4276
weighted avg	0.64	0.04	0.08	4276

Confusion Matrix:

```
[[ 51   7   0  20  541   0   0]
 [ 12 104  13   5 1026   0   0]
 [   5   7  19   1  559   0   0]
 [   3   0   3   3  521   0   0]
 [   0   0   0   0   0   0   0]
 [   2   0   0   1  234   0   0]
 [  16  21   1   3 1088   0  10]]
```

Out[15]: =====

Loading model: phi35_mini_instruct -> microsoft/Phi-4-mini-instruct

Out[15]:

config.json: 0.00B [00:00, ?B/s]

Out[15]:

tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]:

tokenizer.json: 0%| | 0.00/15.5M [00:00<?, ?B/s]

Out[15]:

added_tokens.json: 0%| | 0.00/249 [00:00<?, ?B/s]

Out[15]:

special_tokens_map.json: 0%| | 0.00/587 [00:00<?, ?B/s]

Out[15]:

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Out[15]:

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Out[15]:

Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Out[15]:

Loading weights: 0%| | 0/194 [00:00<?, ?it/s]

Out[15]:

generation_config.json: 0%| | 0.00/168 [00:00<?, ?B/s]

phi35_mini_instruct: 100%|██████████| 4276/4276 [18:05<00:00, 3.94it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:

Out[15]: y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.32296538821328347
Macro F1: 0.2987579547200808
Micro F1: 0.32296538821328347
Weighted F1: 0.39201344456466997
Macro Precision: 0.4168883898439744
Macro Recall: 0.2611918082539424
Balanced Accuracy: 0.3047237762962661
Matthews Corrcoeff: 0.27060592955381474
Cohen Kappa: 0.2389782664986959

Classification Report:

	precision	recall	f1-score	support
Analyze	0.48	0.22	0.31	619
Apply	0.78	0.25	0.38	1160
Create	0.37	0.51	0.43	591
Evaluate	0.43	0.45	0.44	530
Invalid	0.00	0.00	0.00	0
Remember	0.11	0.04	0.06	237
Understand	0.75	0.35	0.48	1139
accuracy			0.32	4276
macro avg	0.42	0.26	0.30	4276
weighted avg	0.59	0.32	0.39	4276

Confusion Matrix:

```
[[139  28  60 116 249   6  21]
 [ 59 295 234  65 420  35  52]
 [   8  13 302  19 228  11  10]
 [  20   9  67 240 174   2  18]
 [   0   0   0   0   0   0   0]
 [  12  10  31  18 126   9  31]
 [  50  24 124  99 425  21 396]]
```

Out[15]: =====
Loading model: mistral_nemo_instruct -> mistralai/Mistral-Nemo-Instruct-2407

Out[15]:
config.json: 0%|██████████| 0.00/622 [00:00<?, ?B/s]

Out[15]:
tokenizer_config.json: 0.00B [00:00, ?B/s]

Out[15]:
tokenizer.json: 0.00B [00:00, ?B/s]

Out[15]:
special_tokens_map.json: 0%|██████████| 0.00/414 [00:00<?, ?B/s]

The tokenizer you are loading from 'mistralai/Mistral-Nemo-Instruct-2407' with an incorrect regex pattern: <https://huggingface.co/mistralai/Mistral-Small-3.1-24B-Instruct-2503/discussions/84#69121093e8b480e709447d5e>. This will lead to incorrect tokenization. You should set the `fix\_mistral\_regex=True` flag when loading this tokenizer to fix this issue.

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Fetching 5 files: 0%| | 0/5 [00:00<?, ?it/s]

Loading weights: 0%| | 0/363 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/116 [00:00<?, ?B/s]

mistral_nemo_instruct: 100%|██████████| 4276/4276 [21:51<00:00, 3.26it/s]
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")

Accuracy: 0.31524789522918617
Macro F1: 0.28129082179156417
Micro F1: 0.31524789522918617
Weighted F1: 0.3456612096231636
Macro Precision: 0.4753228858181818
Macro Recall: 0.24591383436280484
Balanced Accuracy: 0.2868994734232723
Matthews Corrcoeff: 0.22039265008024228
Cohen Kappa: 0.19475546767989238

Classification Report:

	precision	recall	f1-score	support
Analyze	0.59	0.16	0.25	619
Apply	0.40	0.60	0.48	1160
Create	0.66	0.10	0.18	591
Evaluate	0.49	0.39	0.43	530
Invalid	0.00	0.00	0.00	0
Remember	0.37	0.28	0.32	237
Understand	0.82	0.19	0.31	1139
accuracy			0.32	4276
macro avg	0.48	0.25	0.28	4276
weighted avg	0.58	0.32	0.35	4276

Confusion Matrix:

```
[ [ 96 184  2  73 254  5  5]  
  [ 21 699 18  53 319 28 22]
```

```
[ 14 295 60 15 189 10 8]
[ 11 117 6 207 182 4 3]
[ 0 0 0 0 0 0 0]
[ 1 88 0 14 58 66 10]
[ 21 348 5 63 417 65 220]]
```

```
In [16]: for model_name, df in all_predictions.items():
          print("=" * 80)
          print("Model:", model_name)
          print(df["prediction"].value_counts(dropna=False))
```

```
Out[16]: =====
Model: mistral_7b_instruct_v02
prediction
Analyze      2374
Apply        629
Evaluate     445
Understand   384
Create       314
Invalid       87
Remember     43
Name: count, dtype: int64
=====
Model: llama3_8b_instruct
prediction
Invalid      1847
Apply        1090
Analyze      697
Evaluate     331
Create       219
Understand   71
Remember     21
Name: count, dtype: int64
=====
Model: phi35_mini_4k_instruct
prediction
Apply        1127
Analyze      987
Understand   924
Evaluate     775
Create       385
Remember     68
Invalid      10
Name: count, dtype: int64
=====
Model: qwen2_7b_instruct
prediction
Analyze      1317
Apply        980
Create       692
Evaluate     624
Understand   457
Invalid      133
```

```

Remember      73
Name: count, dtype: int64
=====
Model: llama31_8b_instruct
prediction
Invalid      1847
Apply        1090
Analyze       697
Evaluate      331
Create        219
Understand    71
Remember      21
Name: count, dtype: int64
=====
Model: qwen25_7b_instruct
prediction
Apply         1085
Understand     987
Evaluate       853
Create         734
Analyze        454
Remember       157
Invalid         6
Name: count, dtype: int64
=====
Model: qwen25_3b_instruct
prediction
Analyze       1852
Evaluate      1332
Apply         570
Create        351
Invalid       109
Understand     52
Remember       10
Name: count, dtype: int64
=====
Model: qwen25_1_5b_instruct
prediction
Apply         1284
Analyze       1188
Understand    1119
Create        262
Invalid       261
Evaluate      161
Remember       1
Name: count, dtype: int64
=====
Model: smollm2_1_7b_instruct
prediction
Invalid       3236
Remember      983
Evaluate       22
Apply         17
Create         8
Understand     7

```

```

Analyze
Name: count, dtype: int64
=====
Model: gemma2_2b_instruct
prediction
Invalid      2190
Apply        672
Analyze      532
Evaluate     319
Understand   289
Create       264
Remember     10
Name: count, dtype: int64
=====
Model: gemma2_9b_instruct
prediction
Invalid      3969
Apply        139
Analyze      89
Create       36
Evaluate     33
Understand   10
Name: count, dtype: int64
=====
Model: phi35_mini_instruct
prediction
Invalid      1622
Create       818
Evaluate     557
Understand   528
Apply        379
Analyze      288
Remember     84
Name: count, dtype: int64
=====
Model: mistral_nemo_instruct
prediction
Apply        1731
Invalid      1419
Evaluate     425
Understand   268
Remember     178
Analyze      164
Create       91
Name: count, dtype: int64

```

In [17]:

```

results_summary_df = pd.DataFrame(all_results)
print(results_summary_df)
results_summary_df.to_csv("all_models_summary.csv", index=False)

```

Out[17]:

	model_name	model_id	accuracy \
0	mistral_7b_instruct_v02	mistralai/Mistral-7B-Instruct-v0.3	0.436623
1	llama3_8b_instruct	meta-llama/Llama-3.1-8B-Instruct	0.175631

2	phi35_mini_4k_instruct	microsoft/Phi-3.5-mini-instruct	0.603835
3	qwen2_7b_instruct	Qwen/Qwen2-7B-Instruct	0.531805
4	llama31_8b_instruct	meta-llama/Llama-3.1-8B-Instruct	0.175631
5	qwen25_7b_instruct	Qwen/Qwen2.5-7B-Instruct	0.611553
6	qwen25_3b_instruct	Qwen/Qwen2.5-3B-Instruct	0.323433
7	qwen25_1_5b_instruct	Qwen/Qwen2.5-1.5B-Instruct	0.456735
8	smollm2_1_7b_instruct	HuggingFaceTB/SmolLM2-1.7B-Instruct	0.022919
9	gemma2_2b_instruct	google/gemma-2-2b-it	0.239710
10	gemma2_9b_instruct	google/gemma-2-9b-it	0.043732
11	phi35_mini_instruct	microsoft/Phi-4-mini-instruct	0.322965
12	mistral_nemo_instruct	mistralai/Mistral-Nemo-Instruct-2407	0.315248

	macro_f1	micro_f1	weighted_f1	macro_precision	macro_recall	\
0	0.352145	0.436623	0.459888	0.496062	0.358136	
1	0.153002	0.175631	0.201325	0.283747	0.131451	
2	0.457689	0.603835	0.601680	0.493658	0.466069	
3	0.430463	0.531805	0.545616	0.503819	0.436146	
4	0.153002	0.175631	0.201325	0.283747	0.131451	
5	0.482904	0.611553	0.612862	0.497745	0.488341	
6	0.229744	0.323433	0.290228	0.388499	0.285231	
7	0.306240	0.456735	0.439519	0.359944	0.317044	
8	0.024847	0.022919	0.015923	0.238604	0.050849	
9	0.241284	0.239710	0.313133	0.418496	0.187439	
10	0.056123	0.043732	0.078628	0.419989	0.031234	
11	0.298758	0.322965	0.392013	0.416888	0.261192	
12	0.281291	0.315248	0.345661	0.475323	0.245914	

	balanced_accuracy	mcc	cohen_kappa
0	0.417825	0.374849	0.323025
1	0.153359	0.081606	0.069438
2	0.543747	0.511346	0.505866
3	0.508837	0.446931	0.431368
4	0.153359	0.081606	0.069438
5	0.569731	0.520064	0.516751
6	0.332769	0.230920	0.201987
7	0.369885	0.325071	0.317047
8	0.059324	0.014032	0.007783
9	0.218679	0.193952	0.158453
10	0.036439	0.088303	0.029589
11	0.304724	0.270606	0.238978
12	0.286899	0.220393	0.194755

Model Evaluation Metrics

We compute:

- Accuracy
- Macro F1 Score
- Precision Score
- Recall Score
- Confusion Matrix

- Balanced Accuracy Score
- Mathews Correlation
- Cohen's Kappa Score

In [18]:

```
from sklearn.metrics import (
    accuracy_score,
    f1_score,
    precision_score,
    recall_score,
    confusion_matrix,
    classification_report,
    balanced_accuracy_score,
    matthews_corrcoef,
    cohen_kappa_score
)

# remove rows with missing labels
result_df = result_df.dropna(subset=["label", "prediction"])

# clean labels
y_true = result_df["label"].astype(str).str.strip()
y_pred = result_df["prediction"].astype(str).str.strip()

# basic metrics
accuracy = accuracy_score(y_true, y_pred)

macro_f1 = f1_score(y_true, y_pred, average="macro", zero_division=0)
micro_f1 = f1_score(y_true, y_pred, average="micro", zero_division=0)
weighted_f1 = f1_score(y_true, y_pred, average="weighted", zero_division=0)

macro_precision = precision_score(y_true, y_pred, average="macro", zero_division=0)
macro_recall = recall_score(y_true, y_pred, average="macro", zero_division=0)

balanced_acc = balanced_accuracy_score(y_true, y_pred)

# additional research metrics
mcc = matthews_corrcoef(y_true, y_pred)
kappa = cohen_kappa_score(y_true, y_pred)

# confusion matrix
cm = confusion_matrix(y_true, y_pred)

print("Accuracy:", accuracy)
print("Macro F1:", macro_f1)
print("Micro F1:", micro_f1)
print("Weighted F1:", weighted_f1)

print("Macro Precision:", macro_precision)
print("Macro Recall:", macro_recall)

print("Balanced Accuracy:", balanced_acc)

print("Matthews Corrcoef:", mcc)
```

```
print("Cohen Kappa:", kappa)

print("\nClassification Report:")
print(classification_report(y_true, y_pred, zero_division=0))

print("\nConfusion Matrix:")
print(cm)
```

Out[18]:

```
Accuracy: 0.31524789522918617
Macro F1: 0.28129082179156417
Micro F1: 0.31524789522918617
Weighted F1: 0.3456612096231636
Macro Precision: 0.4753228858181818
Macro Recall: 0.24591383436280484
Balanced Accuracy: 0.2868994734232723
Matthews Corrcoeff: 0.22039265008024228
Cohen Kappa: 0.19475546767989238
```

Classification Report:

	precision	recall	f1-score	support
Analyze	0.59	0.16	0.25	619
Apply	0.40	0.60	0.48	1160
Create	0.66	0.10	0.18	591
Evaluate	0.49	0.39	0.43	530
Invalid	0.00	0.00	0.00	0
Remember	0.37	0.28	0.32	237
Understand	0.82	0.19	0.31	1139
accuracy			0.32	4276
macro avg	0.48	0.25	0.28	4276
weighted avg	0.58	0.32	0.35	4276

Confusion Matrix:

```
[[ 96 184   2  73 254   5   5]
 [ 21 699  18  53 319  28  22]
 [ 14 295  60  15 189  10   8]
 [ 11 117   6 207 182   4   3]
 [  0   0   0   0   0   0   0]
 [  1  88   0  14  58  66  10]
 [ 21 348   5  63 417  65 220]]
```

Out[18]:

```
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:2924: UserWarning:
y_pred contains classes not in y_true
warnings.warn("y_pred contains classes not in y_true")
```

Save Predictions

In [19]:

```
result_df = test_df.copy()
result_df["model_name"] = model_name
```

```

result_df["model_id"] = model_id
result_df["prediction"] = predictions

result_df.to_csv(f"{model_name}_predictions.csv", index=False)

y_true = result_df["label"].astype(str).str.strip()
y_pred = result_df["prediction"].astype(str).str.strip()

```

```

In [2]: # Saves the HTML in a specific folder
!jupyter nbconvert --to html notebook.ipynb --output-dir="C:/Users/Reports"

# To specify both a custom folder AND a new filename
!jupyter nbconvert --to html notebook.ipynb --output="final_report.html" --output-dir="C:/Users/

```

```

Out[2]: WARNING: Running pip as the 'root' user can result in broken permissions and conflicting
behaviour with the system package manager, possibly rendering your system unusable. It is
recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv. Use the
--root-user-action option if you know what you are doing and want to suppress this warning.

```

```

Out[2]: -----
ImportError                                Traceback (most recent call last)
Cell In[2], line 4
      1 get_ipython().system('pip install --quiet python-pptx nbformat')
      3 import pathlib, nbformat
----> 4 from pptx import Presentation
      5 from pptx.util import Inches, Pt
      7 def notebook_to_pptx(nb_path: pathlib.Path, out_path: pathlib.Path):
      8     # Load notebook

File /usr/local/lib/python3.11/dist-packages/pptx/__init__.py:9
      6 from typing import TYPE_CHECKING
      8 import pptx.exc as exceptions
----> 9 from pptx.api import Presentation
     10 from pptx.opc.constants import CONTENT_TYPE as CT
     11 from pptx.opc.package import PartFactory

File /usr/local/lib/python3.11/dist-packages/pptx/api.py:14
     11 from typing import IO, TYPE_CHECKING
     13 from pptx.opc.constants import CONTENT_TYPE as CT
---> 14 from pptx.package import Package
     16 if TYPE_CHECKING:
     17     from pptx import presentation

File /usr/local/lib/python3.11/dist-packages/pptx/package.py:8
      5 from typing import IO, Iterator
      7 from pptx.opc.constants import RELATIONSHIP_TYPE as RT
----> 8 from pptx.opc.package import OpcPackage
      9 from pptx.opc.packuri import PackURI
     10 from pptx.parts.coreprops import CorePropertiesPart

File /usr/local/lib/python3.11/dist-packages/pptx/opc/package.py:14
     12 from pptx.opc.constants import RELATIONSHIP_TARGET_MODE as RTM

```

```
13 from pptx.opc.constants import RELATIONSHIP_TYPE as RT
---> 14 from pptx.opc.xml import CT_Relationships, serialize_part_xml
15 from pptx.opc.packuri import CONTENT_TYPES_URI, PACKAGE_URI, PackURI
16 from pptx.opc.serialized import PackageReader, PackageWriter
```

File /usr/local/lib/python3.11/dist-packages/pptx/opc/xml.py:7

```
3 from __future__ import annotations
5 from typing import TYPE_CHECKING, Callable, cast
----> 7 from lxml import etree
9 from pptx.opc.constants import NAMESPACE as NS
10 from pptx.opc.constants import RELATIONSHIP_TARGET_MODE as RTM
```

ImportError: cannot import name 'etree' from 'lxml' (/usr/lib/python3/dist-packages/lxml/__init__)